Principles of Geographic Information Systems

Modeling the distribution of marine biodiversity under global climate change

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What is GIS? Why does it matter for modeling biodiversity?

Definition: a system to capture, store, manipulate, analyze, and present spatial or data.

Relevance in marine science: mapping habitats, tracking species distributions under climate change, analyzing environmental patterns (e.g., temperature anomalies), conservation planning (MPAs).

Specific link to SDM: supports the representation of species locations and environmental variables; crucial for data preparation, interpretation and visualization.

Representing reality: spatial data models

Real-world features are represented digitally by **vector** and **raster data models**.

Vector data model: points, lines, polygons

Represents **discrete features with distinct boundaries**.

Primitives:

- **Points:** x, y (,z) locations (e.g., occurrences, stations).
- **Lines:** points connected by segments (e.g., coastlines, tracks, currents).
- **Polygons:** closed sequence of segments representing areas (e.g., MPAs, islands).

Features are linked to an attribute table (e.g., species name, MPA size).

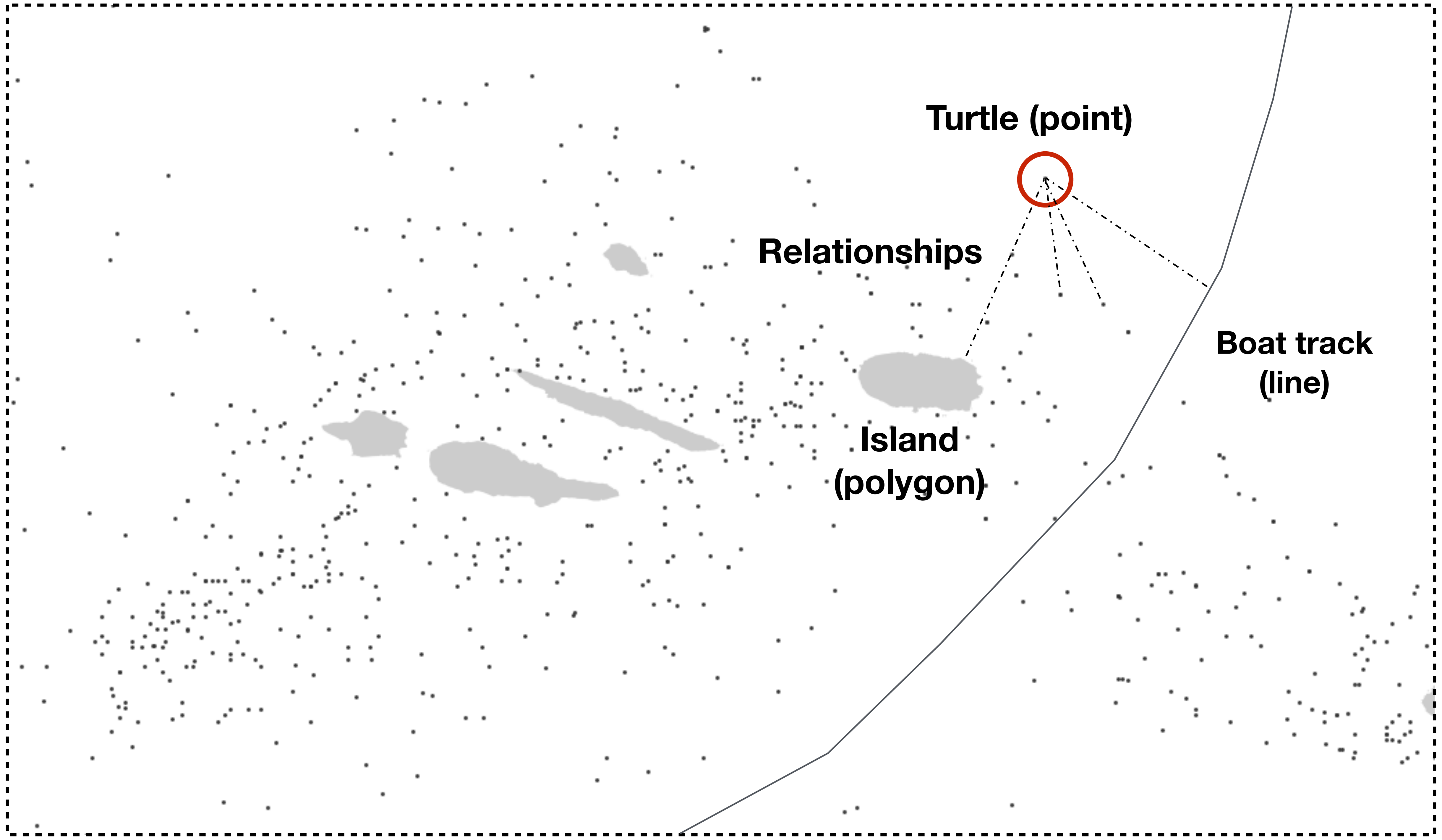
Can store spatial relationships (e.g., distances between sites).

Spatial information for *Caretta caretta*

Locations		Attributes						Relationships
Lon	Lat	Species	Common name	Date	Depth	Behaviour	Observer	Distance to land
0,61287	39,58955	Caretta caretta	Loggerhead sea turtle	10/08/12	22	Swimming	John Doe	232
0,83999	39,77417	Caretta caretta	Loggerhead sea turtle	03/03/12	12	Swimming	John Doe	3454
3,18192	42,40479	Caretta caretta	Loggerhead sea turtle	14/08/11	1	Eating	Unknown	23
-4,86395	55,24295	Caretta caretta	Loggerhead sea turtle	01/01/04	0	Swimming	Unknown	2567
-5,89297	57,06977	Caretta caretta	Loggerhead sea turtle	03/03/12	0	Sleeping	John Doe	878
34,9082	32,55822	Caretta caretta	Loggerhead sea turtle	22/12/02	23	Sleeping	John Doe	777
0,61287	39,58955	Caretta caretta	Loggerhead sea turtle	14/08/11	12	Swimming	John Doe	6
0,83999	39,77417	Caretta caretta	Loggerhead sea turtle	03/03/12	2	Sleeping	John Doe	78

Vector data model

Points, lines and polygons (locations, and features with attributes)



The distribution of the Loggerhead sea turtle in the Azores Islands

Raster data model: grids and cells

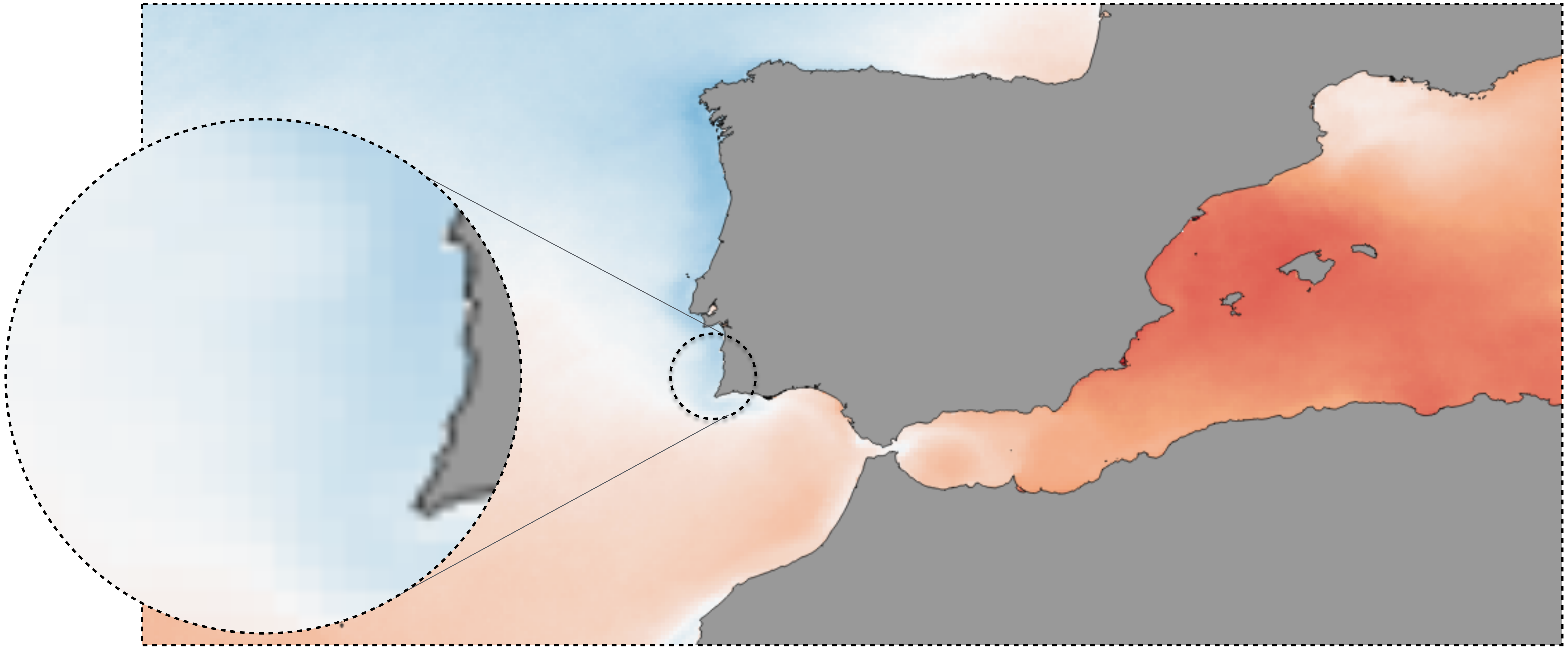
Represents space as a grid of cells (pixels).

Each cell holds a single value (e.g., temperature, depth, suitability score).

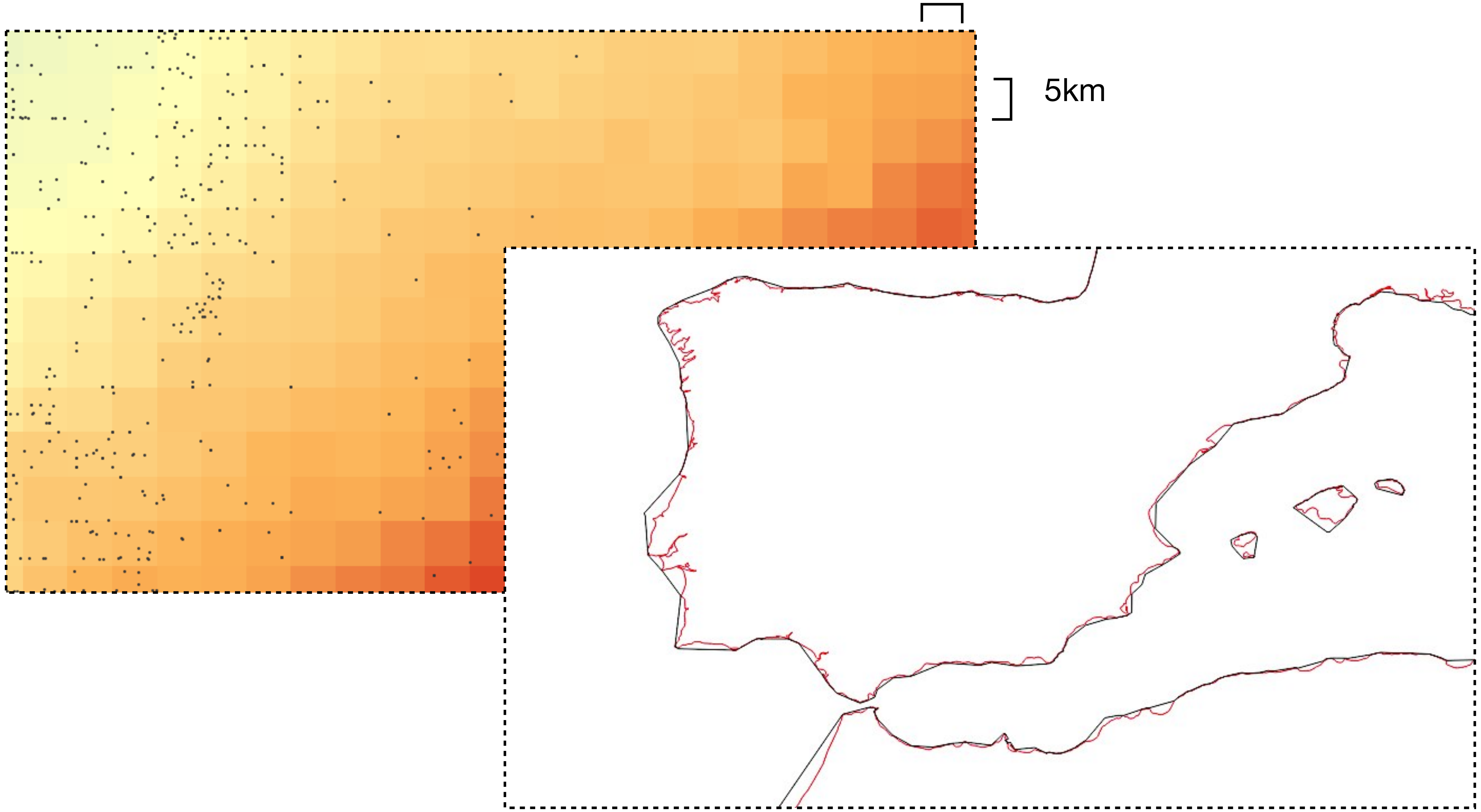
Raster data can be continuous (e.g., temperature) or thematic / categorical (e.g., habitat type).

Raster data model

Grids and cells representing continuous features



SST in the western Mediterranean Sea and adjacent Atlantic Ocean.



Both data models have well-defined **spatial resolutions**, which is the size of the **smallest feature able to be recognized**.

Vector data model: number of vertex;

Raster data model: size of cells;

Key differences between vectors and rasters:

Vector: good for discrete features, with numerous attributes and relationships (attribute tables).

Raster: good for continuous features (one attribute by the cell value).

Uses in SDM: occurrences (vector); Environmental conditions (raster).

Conversion: possible (rasterization, vectorization), but can involve information change or loss.

Where on earth? Coordinate Reference Systems (CRS)

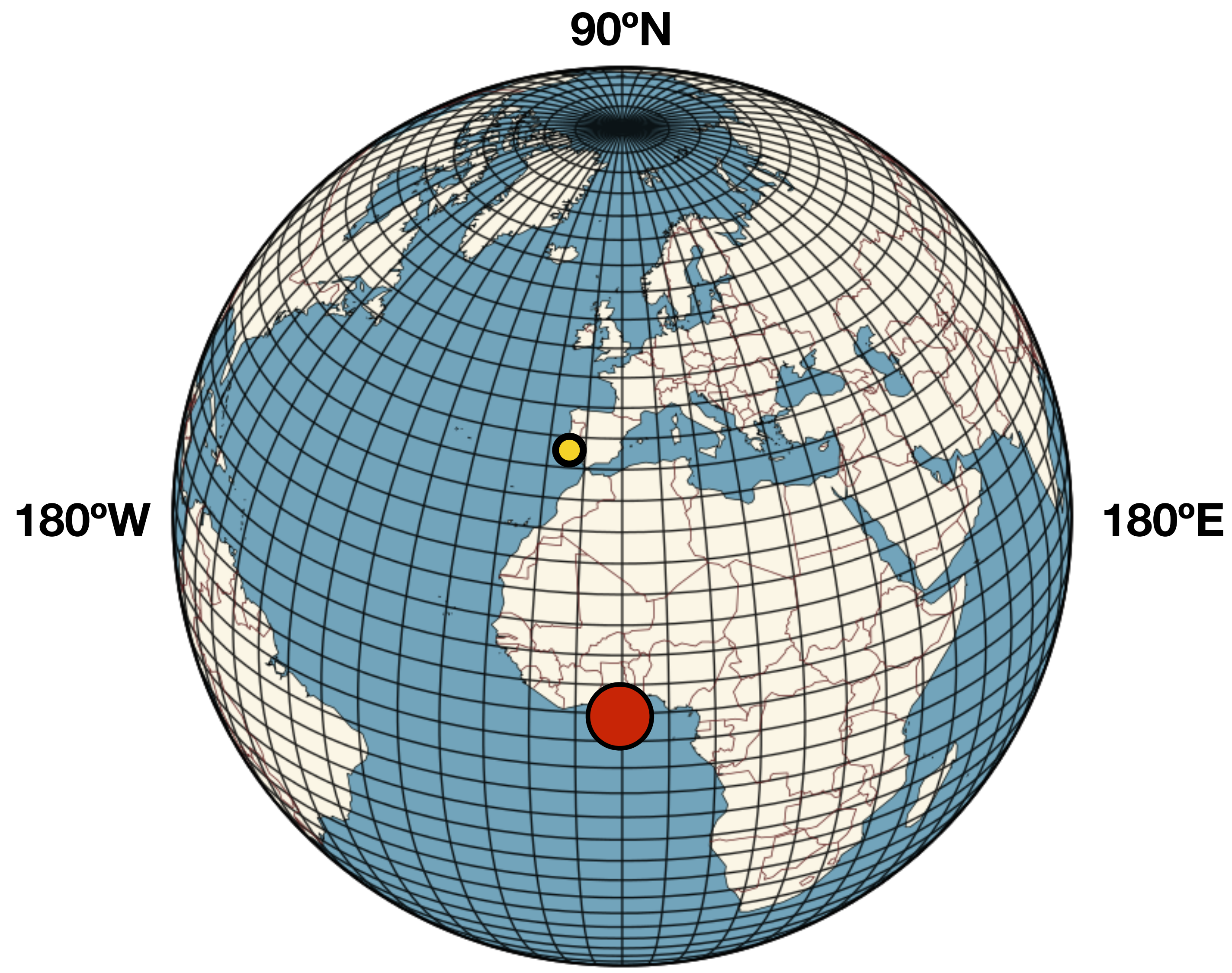
Coordinate reference systems **allow representing the locations of features within a common geographic framework**.

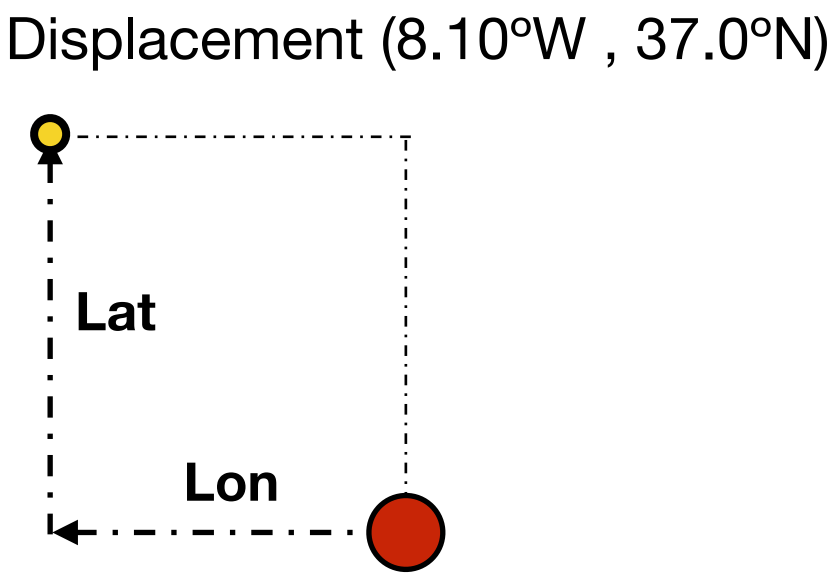
Without a CRS, coordinates are just numbers; they don't represent a specific place.

e.g., 7.321 ; -121.981

A CRS gives meaning to the information of location.

CRS have an inherent fundamental challenge: representing the 3D Earth on 2D surfaces.





Reference point (Datum) at the interception between:

Equator & Prime Meridian (0ºN 0ºE)

CRS type 1: Geographic Coordinate Systems (GCS)

Defines locations directly on the 3D Earth using latitude and longitude (angles).

Components:

- **Datum:** positions locations relative to the Earth.
- **Displacement units:** degrees (usually decimal degrees).

Decimal degrees are simpler to use.

37°2'38.08"N ; 7°58’20.73”W

37.043911° ; -7.972425° 👈🏼

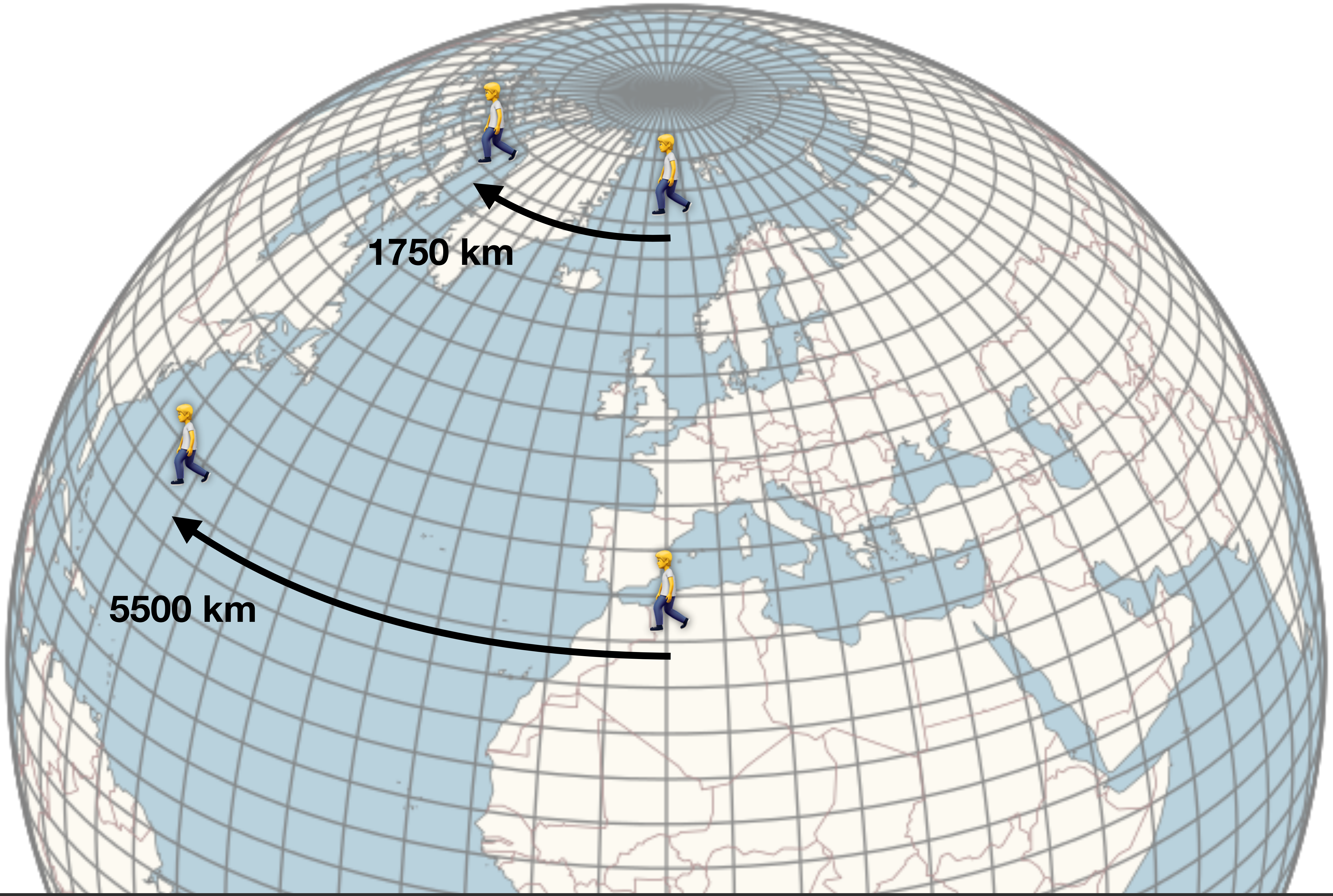
Simpler to type and share with others;

In data tables (e.g., R, Excel), longitude and latitude entries **use single cells, allowing direct analyses** (e.g., calculate distances).

Geographic coordinate systems

GCS **are not good to make measurements**.

There is the need to project data to an alternative coordinate system (e.g., based on actual distances).

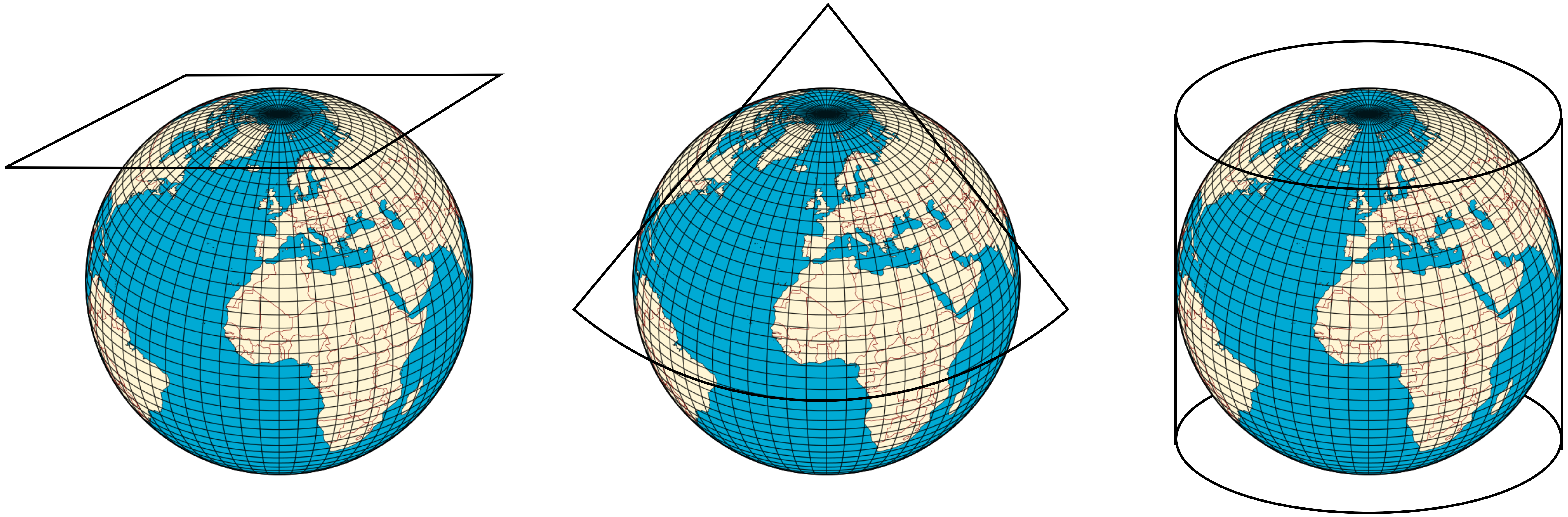


Displacement of 60º performed at different latitudes (360º longitude).

CRS type 2: Projected Coordinate Systems (PCS)

Mathematical transformation (**projection**) of the Earth's surface onto a 2-dimensional plane (**map**) with **constant units of distance (e.g., m), area, direction across the two dimensions**.

Crucially, **all map projections introduce distortion**. It's impossible to perfectly flatten a curved surface. Projections inevitably distort one or more properties.

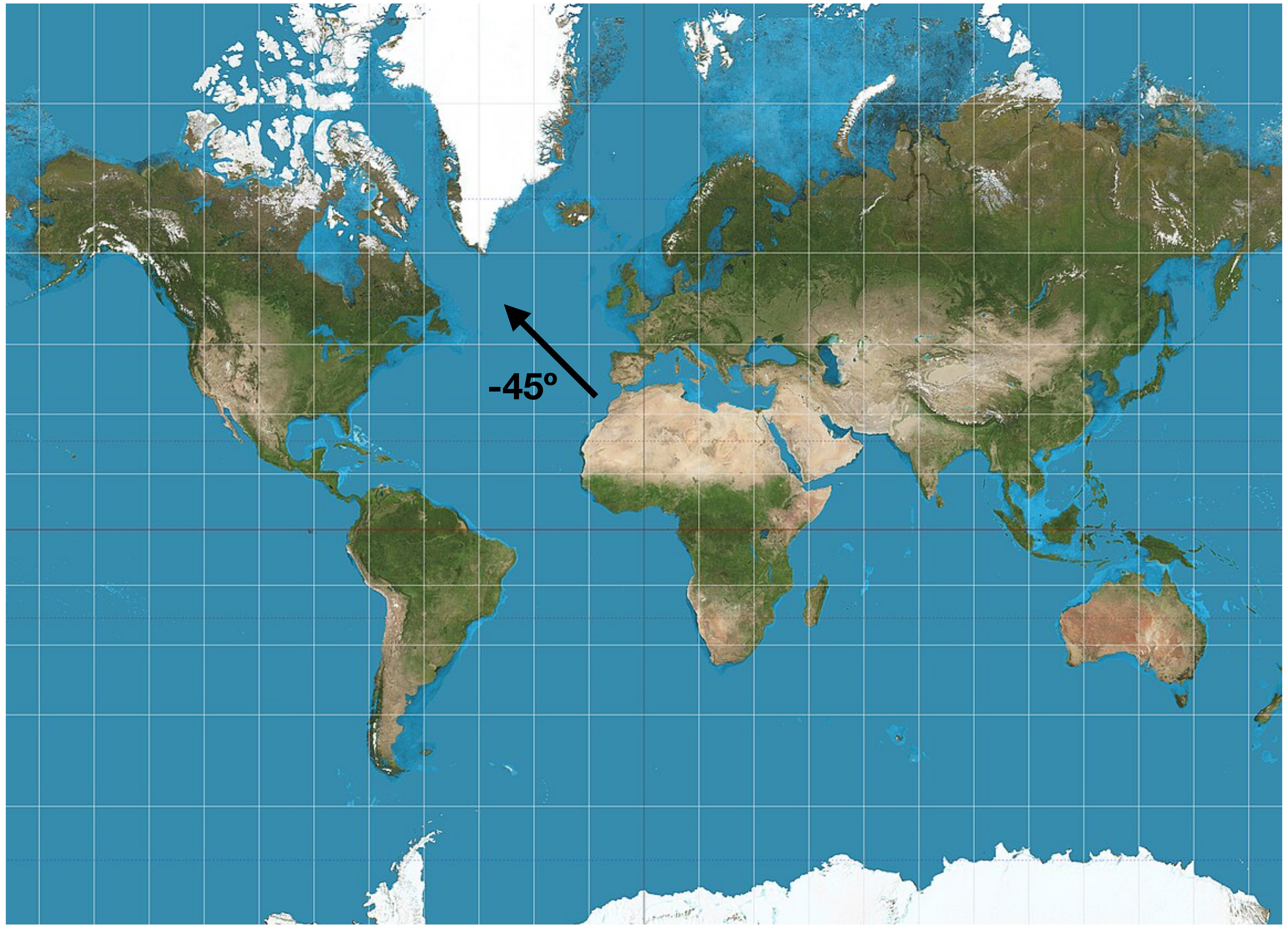


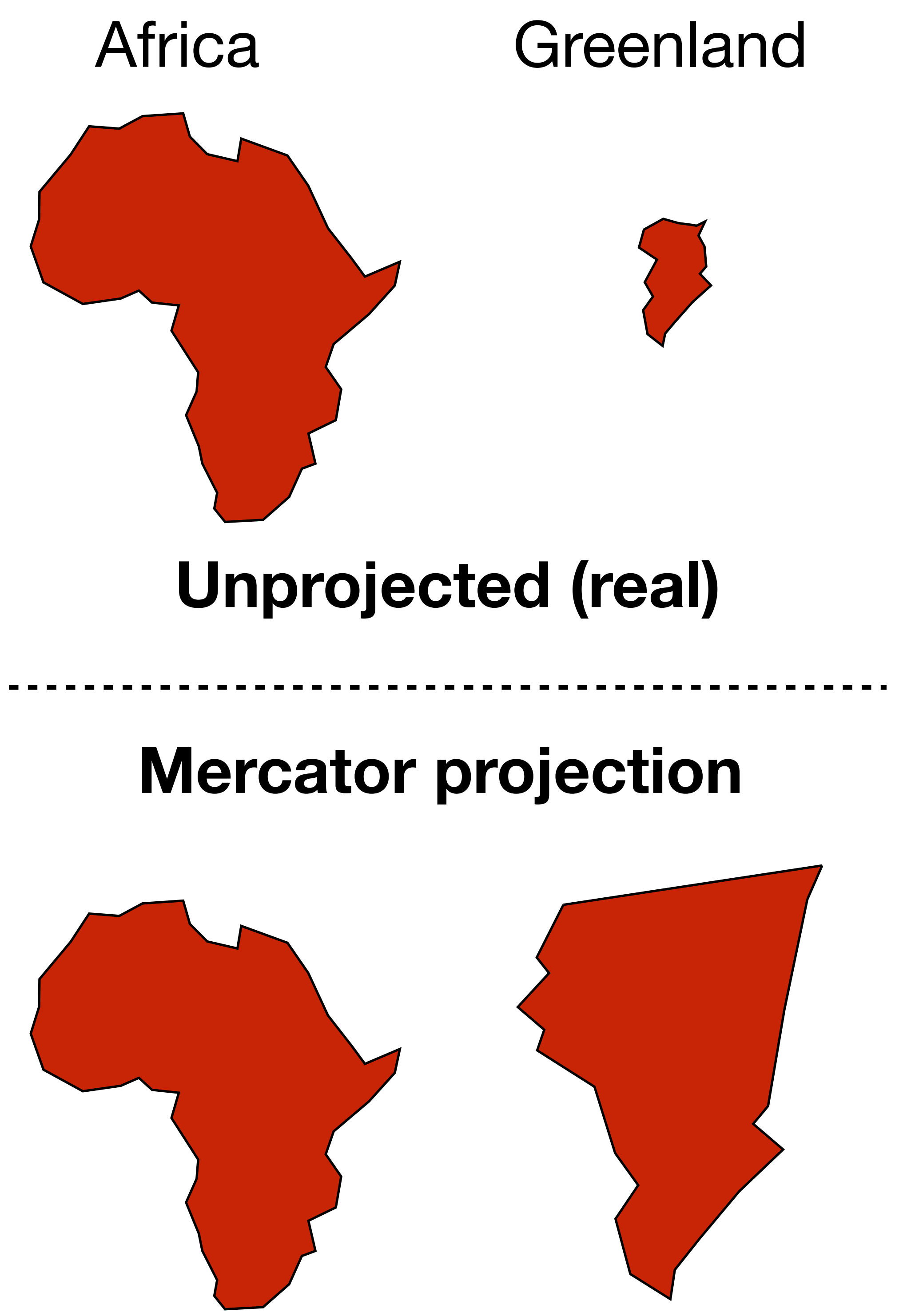
Planar projection (poles)

Conical projection (mid-latitudes)

Cylindrical projection (rectangular maps)

There is no best projection. **Map-maker must decide which property is important** for the application (e.g., equal area for area calculations, equidistant for distance measurements).





Mercator is a cylindrical projected coordinate systems that **represents lines of constant angle** (i.e., course; good for nautical use).

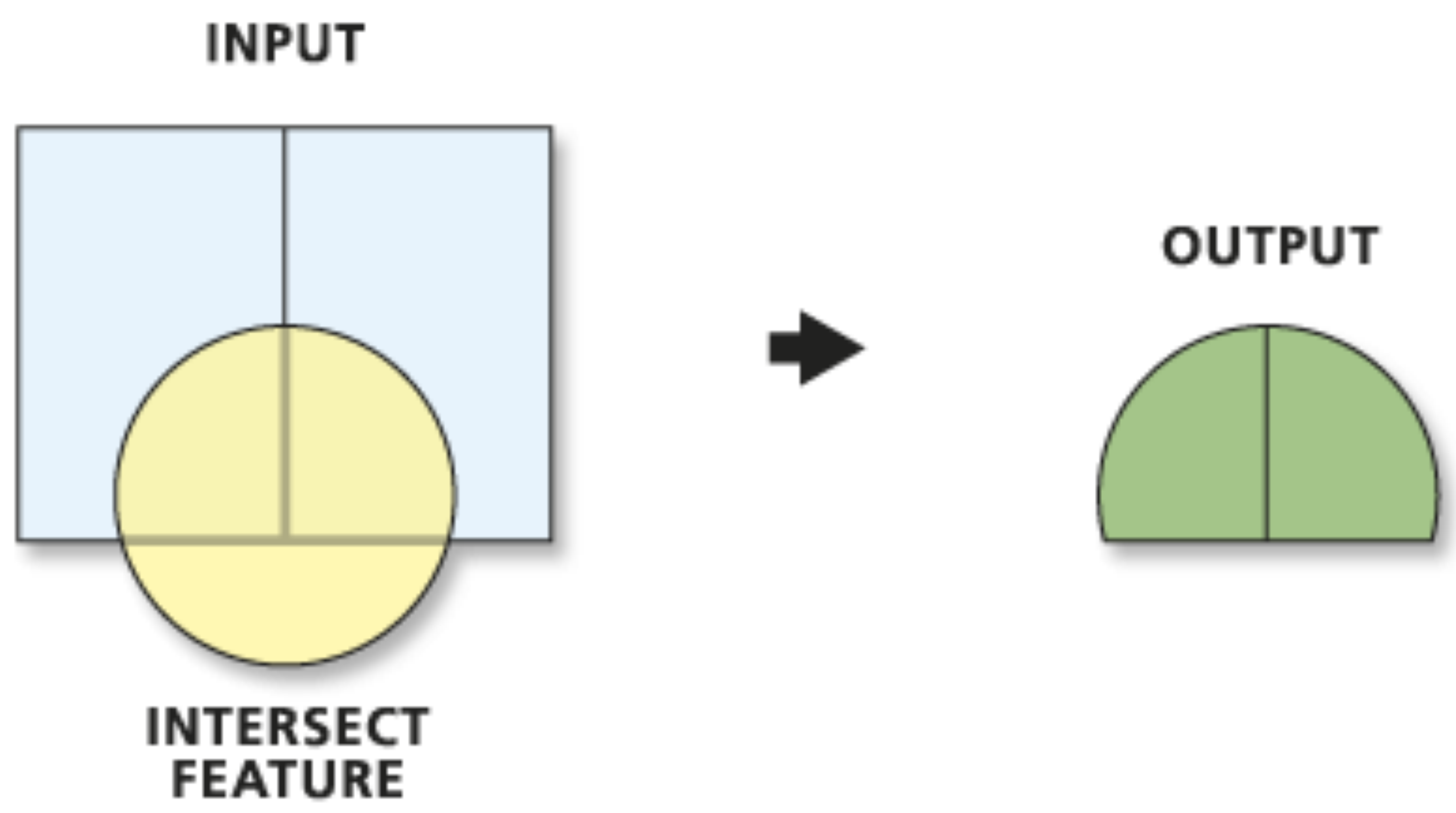
Mercator projection, distortion increases towards higher latitudes.

What can GIS do?

GIS provides powerful analytical tools beyond mapping.

Vector geoprocessing capabilities: Intersection

Intersection extracts the overlapping portions of features in the Input by an Intersect layer. Attributes are not modified.

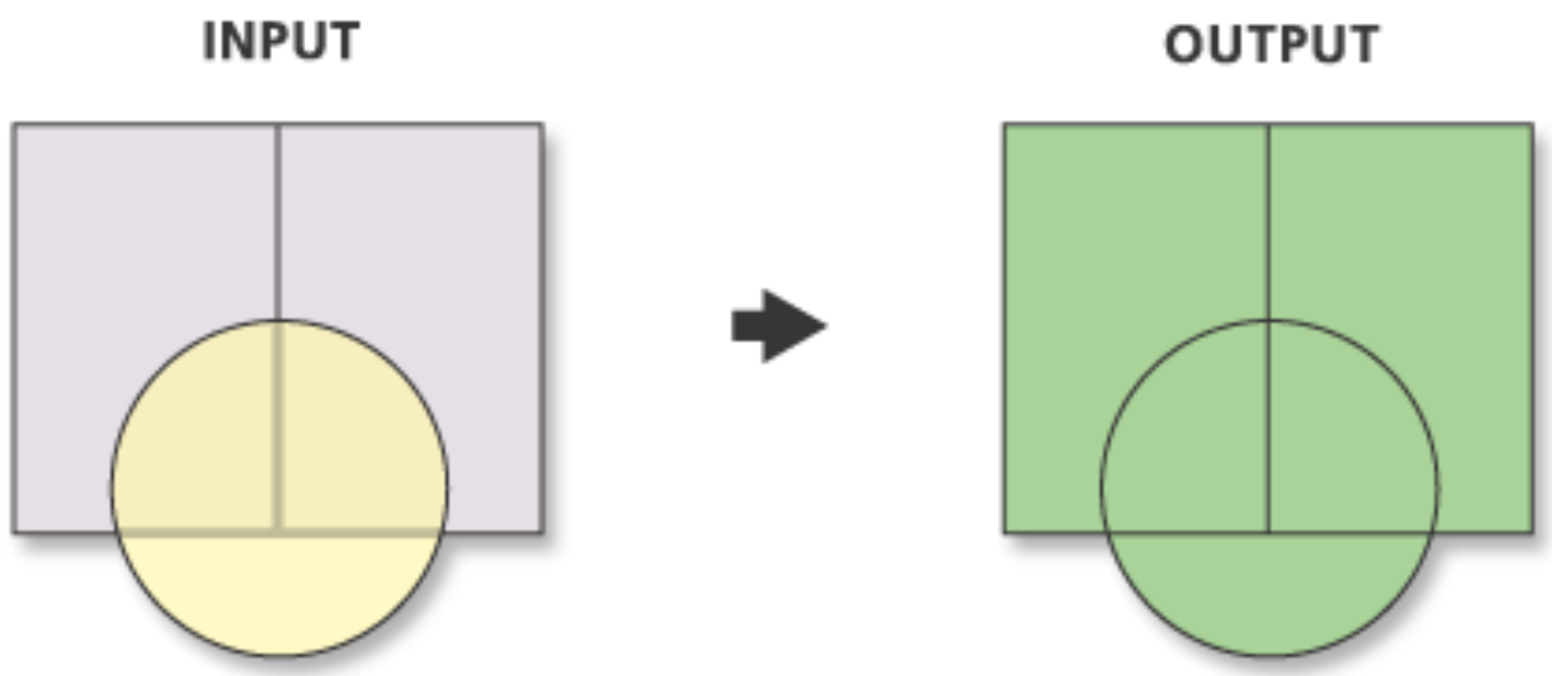


e.g., get the fishing areas for a species (input) within a MPA (intersect)

Vector geoprocessing capabilities: Union

Union completely merges of two features.

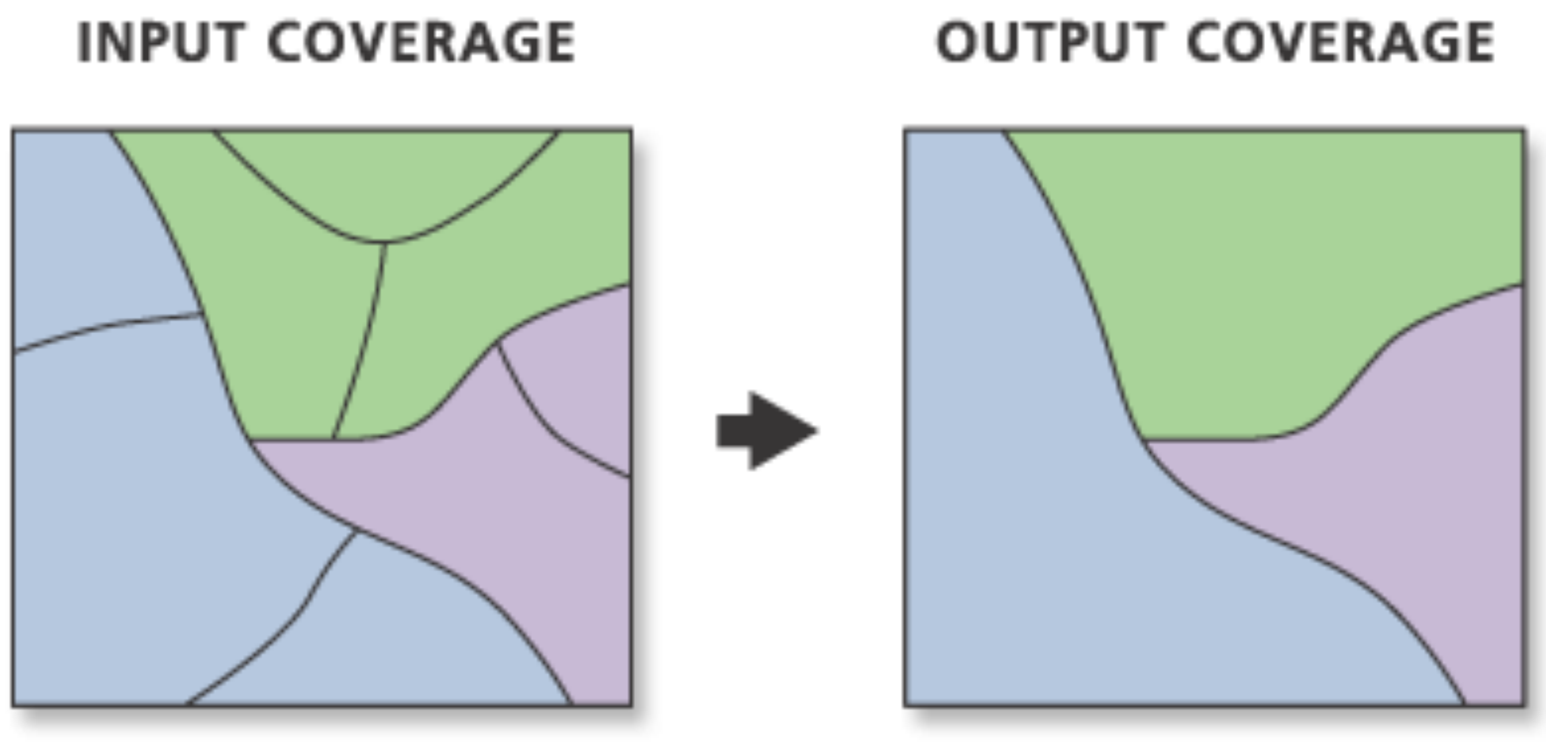
All the fields from both data attribute tables are kept.



e.g., get the distribution of a genus composed by two species (input).

Vector geoprocessing capabilities: Dissolve

Dissolve merges boundaries of adjacent polygons with the same attribute into one larger polygon.



e.g., identify contiguous populations from the same species, to determine or show global areas.

Raster geoprocessing capabilities: Map Algebra

Execute algebra expressions between raster layers to output a new raster.

Expressions include: count, minimum, maximum, sum, mean, standard deviation, …

Max(

1	1	2	**2**
4	**4**	**2**	**2**
4	1	1	2
4	3	3	**6**
1	**4**	1	6

,

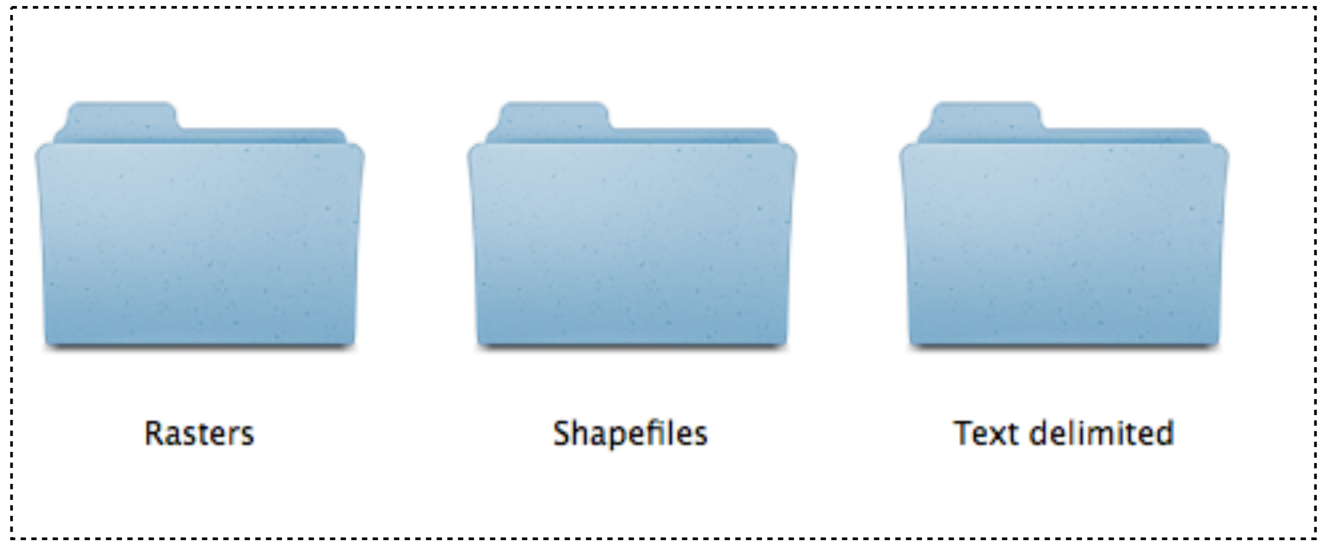
3	**4**	**6**	1
4	3	0	0
1	**4**	**6**	**2**
2	**6**	**3**	6
4	1	**14**	6

) =

3	**4**	**6**	**2**
4	**4**	**2**	**2**
4	**4**	**6**	**2**
4	**6**	**3**	**6**
4	**4**	**14**	**6**

e.g., get the maximum SST between two summer months

Common file formats



Folders with spatial data

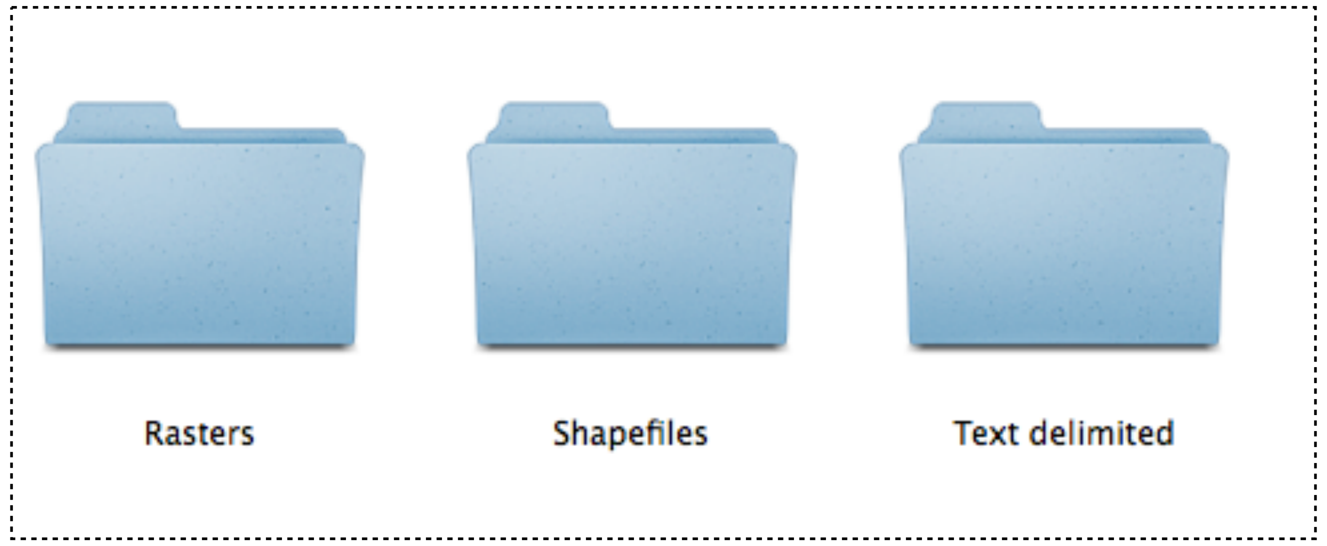
Vector data are common as **shapefiles** (a standard since 1990).

.shp is the geometry of vector features

.dbf are the attributes of vector features

.shx helps the GIS Application to find features more quickly.

Common file formats



Folders with spatial data

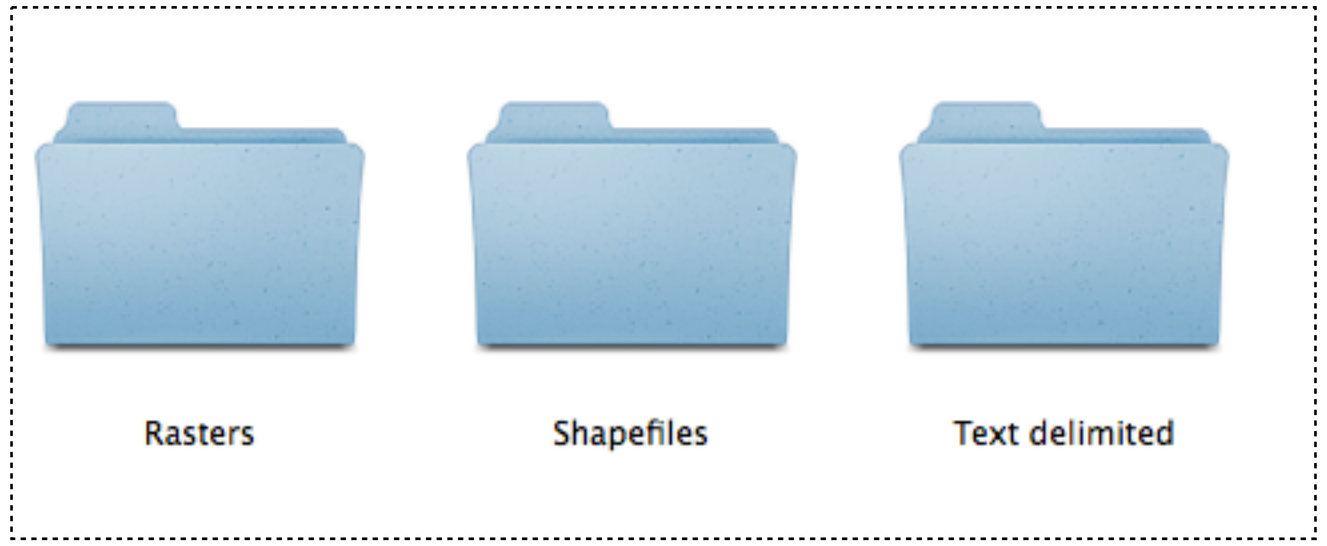
Vector data are migrating to simpler formats like **GeoPackage files**, which are an open, portable, self-describing, compact formats for spatial information.

saccorhiza

ID	SPECIES	Y	LON	LAT	REF
1	Sacchorhiza polyschides	1966	-5.600819	36.018775	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
2	Sacchorhiza polyschides	1993	-5.601822	36.019239	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
3	Sacchorhiza polyschides	1960	-5.653864	36.062617	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
4	Sacchorhiza polyschides	1965	-5.44705	36.116542	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
5	Sacchorhiza polyschides	1958	-5.353586	36.14075	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
6	Sacchorhiza polyschides	1965	-5.922481	36.190028	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
7	Sacchorhiza polyschides	1966	-6.087778	36.277683	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
8	Sacchorhiza polyschides	1983	-5.158386	36.422061	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
9	Sacchorhiza polyschides	1982	-4.717386	36.493706	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
10	Sacchorhiza polyschides	1982	-4.882289	36.510117	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
11	Sacchorhiza polyschides	1982	-4.622617	36.571558	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
12	Sacchorhiza polyschides	1966	-4.421261	36.721283	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
13	Sacchorhiza polyschides	1970	-8.936489	37.009864	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
14	Sacchorhiza polyschides	1963	-8.978044	37.023033	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
15	Sacchorhiza polyschides	1963	-8.667864	37.079811	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
16	Sacchorhiza polyschides	1963	-8.266667	37.083333	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
17	Sacchorhiza polyschides	1956	-8.45	37.083333	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX. Saccor
18	Sacchorhiza polyschides	1970	-8.247875	37.089072	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
19	Sacchorhiza polyschides	1970	-8.469325	37.098511	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
20	Sacchorhiza polyschides	1970	-8.896	37.184133	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX. Saccor
21	Sacchorhiza polyschides	1954	-8.770147	37.757611	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
22	Sacchorhiza polyschides	1970	-8.796936	37.906147	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
23	Sacchorhiza polyschides	1970	-9.216664	38.416647	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
24	Sacchorhiza polyschides	1979	-9.1176	38.43685	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX. Saccor
25	Sacchorhiza polyschides	1970	-9.101489	38.444469	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
26	Sacchorhiza polyschides	1989	-8.984847	38.469364	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
27	Sacchorhiza polyschides	1982	-8.990794	38.495931	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
28	Sacchorhiza polyschides	1970	-9.355975	38.689786	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
29	Sacchorhiza polyschides	1989	-9.483333	38.7	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX. Saccor
30	Sacchorhiza polyschides	1951	-9.269286	38.702911	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX

Vector data are also available as **text delimited files** (e.g., .txt, .csv), delimited by a **Tab, a comma or semicolon**. Useful for point data and matrices (e.g., excel).

Common file formats



Folders with spatial data

Raster data are the the industry standard for GIS and satellite remote sensing applications. **Files are typically as .tif, .asc or .nc**.